

Derivations

Problem - 1

A.1) Let the Earth's magnetic field in the body frame be

$$B_e = BU, \quad B = \|B_e\|, \quad \|U\| = 1$$

where

$$U = [U_x \ U_y \ U_z]^T$$

contains the direction cosines of the Earth field.

Let B_p be the field generated by the platform.

A scalar magnetometer measures $m = \|B_e + B_p\|$

$$\begin{aligned} m^2 &= (B_e + B_p)^T (B_e + B_p) \\ &= B^2 + 2B_e^T B_p + \|B_p\|^2 \end{aligned}$$

Since $B_e = BU$

$$m = B \sqrt{1 + \frac{2U^T B_p}{B} + \frac{\|B_p\|^2}{B^2}}$$

Condition/Assumption $\frac{\|B_p\|}{B} \ll 1$

In this case, we can use the Taylor expansion:

$$\sqrt{1+x} \approx 1 + \frac{x}{2} \text{ and neglect higher order terms.}^*$$

$$m \approx B + U^T B_p$$

$\therefore$ the first order scalar interference is:

$$\Delta m = m - B \approx U^T B_p$$

The scalar magnetometer only sees the component of platform field parallel to Earth field in first order.

A.2 Permanent magnetisation

$$B_{perm} = a = [a_x \ a_y \ a_z]^T$$

$$\Delta m_{perm} = U^T a = a_x U_x + a_y U_y + a_z U_z$$

A.3 Induced magnetisation

for soft magnetic material with linear response to field,

$$B_{ind} = \mu B_e = \mu \mu_0 H$$

where M is a fixed 3×3 susceptibility tensor

$$\therefore \Delta m_{\text{ind}} = U^T B_{\text{ind}} = B U^T M U$$

decomposing into symmetric and asymmetric

$$M = S + A$$

$$S = \frac{M + M^T}{2}, \quad A = \frac{M - M^T}{2}$$

$$U^T A U = (U^T A U)^T = U^T A^T U = -U^T A U \quad (\text{since } A^T = -A)$$

$\therefore$ only symmetric part of M is observable

$$S = \begin{bmatrix} \beta_4 & \beta_5/2 & \beta_6/2 \\ \beta_5/2 & \beta_7 & \beta_8/2 \\ \beta_6/2 & \beta_8/2 & \beta_9 \end{bmatrix}$$

$$\left. \begin{array}{l} \beta_1 = a_x \\ \beta_2 = a_y \\ \beta_3 = a_z \end{array} \right\} \text{from permanent}$$

$$\therefore \Delta m_{\text{ind}} = B(\beta_4 U_x^2 + \beta_5 U_x U_y + \beta_6 U_x U_z + \beta_7 U_y^2 + \beta_8 U_y U_z + \beta_9 U_z^2)$$

A.4 Eddy currents

$$B_{\text{eddy}} = C B_e$$

with $C \in \mathbb{R}^{3 \times 3}$

$$B_e = B U + B U$$

Assumption: during calibration, the changes caused by attitude are faster than changes in field magnitude:

$$|\dot{B}| \ll B \|\dot{U}\|$$

therefore, $\dot{B}_c \approx B \dot{U}$

$$\therefore B_{\text{eddy}} \approx B C \dot{U}$$

$$\Delta m_{\text{eddy}} \approx B U^T C \dot{U}$$

with

$$C = \begin{bmatrix} \beta_0 & \beta_{11} & \beta_{12} \\ \beta_{13} & \beta_{14} & \beta_{15} \\ \beta_{16} & \beta_{17} & \beta_{18} \end{bmatrix}$$

Combining the three mechanisms, we get that standard 18-term Tolke-Lawson model:

$$\begin{aligned} \Delta m = & \beta_1 U_x + \beta_2 U_y + \beta_3 U_z \\ & + B(\beta_4 U_x^2 + \beta_5 U_x U_y + \beta_6 U_x U_z + \beta_7 U_y^2 \\ & + \beta_8 U_y U_z + \beta_9 U_z^2) \\ & + B(\beta_{10} U_x \dot{U}_x + \beta_{11} U_x \dot{U}_y + \dots + \beta_{18} U_z \dot{U}_z) \end{aligned}$$

B.1 Since u is a unit vector $u^T u = 1$

$$\frac{d}{dt}(u^T u) = 2u^T \dot{u} = 0$$

$$\therefore u_x \dot{u}_x + u_y \dot{u}_y + u_z \dot{u}_z = 0$$

Defining $\phi(t)$ with 18 Tollen-Lauzon regressors,

$$\phi(t) = [u_x \ u_y \ \dots \ Bu_x^2 \ \dots \ Bu_z \dot{u}_z]$$

$$\text{Let } \beta = [\beta_1 \ \dots \ \beta_{18}]^T$$

$$\Delta m(t) = \phi(t)\beta$$

$$\text{for } N \text{ measurements: } A = \begin{bmatrix} \phi(t_1) \\ \phi(t_2) \\ \vdots \\ \phi(t_N) \end{bmatrix} \in \mathbb{R}^{N \times 18}$$

$$y = A\beta + n$$

$$\text{Since } u_x \dot{u}_x + u_y \dot{u}_y + u_z \dot{u}_z = 0$$

Columns of A with these terms

$$A_{10} + A_{14} + A_{18} = 0$$

therefore, $\text{rank}(A) \leq 17$

B.2

Let's take a linear combination of all regressors
that vanishes for every physically
allowed

$$\|u\| = 1, \quad u^T \dot{u} = 0$$

$$a^T u + b u^T S u + b u^T C \dot{u} = 0$$

$$S = S^T$$

Let's choose $\dot{u} = 0$
Then,

$$a^T u + b u^T S u = 0$$

replacing $u \rightarrow -u$

$$-a^T u + b u^T S u = 0$$

$$\Rightarrow 2a^T u = 0 \quad \forall u$$

$$\therefore a = 0$$

$$\text{Then } u^T S u = 0 \quad \forall u$$

$\therefore S = 0$ must be satisfied.

Therefore any remaining null direction must
come from the eddy term:

$$v^T C v = 0$$

for every $v \perp U$
for fixed U ,

$$(C^T U)^T v = 0 \quad \forall v \perp U$$

hence $C^T U \parallel U \quad \forall U$

$$\boxed{C = \lambda I}$$

which implies: $v^T C \dot{v} = \lambda v^T \dot{v} = 0$

B.2) Induced sensor M

for a general M , $v^T M v = v^T \left(\frac{M + M^T}{2} \right) v$

the antisymmetric part of M is unobservable

Eddy Tensor

for any λ

$$v^T (C + \lambda I) \dot{v} = v^T C \dot{v} + \lambda \underbrace{v^T \dot{v}}_{=0}$$

hence, C and $C + \lambda I$
produce identical scalar measurements

So only $C - \frac{\text{tr} C}{3} I$ is observable.

$$\text{since } u_x^2 + u_y^2 + u_z^2 = 1$$

The three diagonal induced terms satisfy:

$$b u_x^2 + b u_y^2 + b u_z^2 = b$$

An isotropic induced component λI
contributes λb ,

now, if the absolute Earth field baseline b
is unknown, this cannot be distinguished
from a change in background field.

→ Practically 16 degrees of freedom

B.4) For a straight flight at constant heading,

$$u(t) = u_0$$

$$\dot{u} = 0$$

$$b(t) \simeq b_0$$

all 9 eddy regressors are zero
all remaining permanent and induced regressors

are constant.

$\therefore$ every row of A is identical

$$\text{rank}(A) = 1$$

$$\text{rank}(\text{BPF}(\text{row } A)) = 0 \quad (\text{meaning, band pass filtering})$$

B.5 Minimal calibration manoeuvres

We need persistent excitation: $u(t)$ must explore different orientations, while $v(t)$ must explore independent tangent directions.

Two independent, non-parallel attitude variations:

$$\phi(t) = A_\phi \sin(\omega_\phi t)$$

$$\theta(t) = A_\theta \sin(\omega_\theta t)$$

These make $y(t)$ explore a two dimensional patch of S^2 .

Theoretical minimum: two independent rotations.

C. Ground-level magnetic anomaly:

$$m_0(r), \quad r = (x, y)$$

modelled as a zero-mean stationary
isotropic Gaussian random field with

$$P_0(k) = Ck^{-\beta}, \quad \beta \in [2.5, 4]$$

At altitude h , upward continuation multiplies

Fourier amplitudes by e^{-kh}

$$\therefore P_h(k) = Ck^{-\beta} e^{-2kh}$$

C(a) Cramer-Rao

Suppose

$$z = m_h(r) + n, \quad n \sim N(0, \sigma^2)$$

for a small displacement δr ,

$$m_h(r + \delta r) \simeq m_h(r) + \nabla m_h(r)^T \delta r$$

defining $g = \nabla m_h(r)$

thus $\Delta m \approx g^T \delta r$

The likelihood is: $p(z|r) = \frac{1}{\sqrt{2\pi}\sigma^2} \exp\left[-\frac{(z-m_h(r))^2}{2\sigma^2}\right]$

$\therefore l(r) = -\frac{(z-m_h)^2}{2\sigma^2} + \text{const}$

and

$$\nabla_r l = \frac{z-m_h}{\sigma^2} g$$

The fisher matrix is $J = E[(\nabla l)(\nabla l)^T]$

$$= E\left[\frac{(z-m_h)^2}{\sigma^4}\right] gg^T$$

$$\Rightarrow J = \frac{1}{\sigma^2} gg^T$$

since
 $z-m_h = n$

$$J = \frac{1}{\sigma^2} \begin{bmatrix} g_x^2 & g_x g_y \\ g_x g_y & g_y^2 \end{bmatrix}$$

$$E[n^2] = \sigma^2$$

$$\text{rank}(J) = 1$$

defining $e_{||} = \frac{g}{\|g\|}$, then $Je_{||} = \frac{\|g\|^2}{\sigma^2} e_{||}$

$$J_{e_{\perp}} = 0$$

CRLB in the observation direction:
 $\rightarrow$ Cramer rao...

$$\text{Var}(\hat{r}_{||}) \geq \frac{\sigma^2}{\| \nabla m_h \|^2}$$

$$\sigma_{\text{pos}, ||} \geq \frac{\sigma}{\| \nabla m_h \|}$$

Position recovers the second dimension,
 along a trajectory

$$g_i = \nabla m_h(r_i)$$

independent measurements add Fisher information,

$$J_{\text{traj}} = \frac{1}{\sigma^2} \sum_{i=1}^N g_i g_i^T$$

if every g_i is parallel

$$\text{rank}(J_{\text{traj}}) = 1$$

if at least two sampled gradients are non-parallel
 and span $\mathbb{R}^2$

$$\text{rank}(J_{\text{traj}}) = 2.$$

C(b) Position information depends on $\|\nabla m_h\|$

$$G_h^2 = E \|\nabla m_h\|^2$$

Using parareal: $G_h^2 = \int_{\mathbb{R}^2} \frac{d^2k}{(2\pi)^2} k^2 P_h(k)$

$$G_h^2 = C \int \frac{d^2k}{(2\pi)^2} k^2 P e^{-2kh}$$

by isotropy $d^2k = k dk d\theta$

integrating over θ , $G_h^2 = \frac{C}{2\pi} \int_0^\infty k^{3-\beta} e^{-2kh} dk$

let $s = 4 - \beta$, $a = 2h$

for $h > 0$, $\beta < 4$

$$\int_0^\infty k^{s-1} e^{-ak} dk = \frac{\Gamma(s)}{a^s}$$

$$G_h^2 = \frac{C}{2\pi} \frac{\Gamma(4-\beta)}{(2h)^{4-\beta}}$$

$$\therefore G_h^2 \propto h^{\beta-4}$$

$$\Rightarrow G_h \propto h^{(\beta-4)/2}$$

By isotropy, $E[g_x^2] = E[g_y^2] = \frac{G_h^2}{2}$

$$E[g_x g_y] = 0$$

$$E[J] = \frac{G_h^2}{2\sigma^2} I_2$$

$$\therefore \sigma_{pos} \sim \frac{\sqrt{2} \sigma}{G_h}$$

$$\sigma_{pos}(h) \propto h^{(4-\beta)/2}$$

$$\gamma = \frac{4-\beta}{2}$$

Finite map size and resolution

A real map cannot contain all k .

For a physical map width L ,

$$k_{min} \sim \frac{2\pi}{L}$$

for fixed spacing Δx

$$k_{\max} \sim \frac{\pi}{\Delta x}$$

Therefore, $G_h^2 = \frac{C}{2\pi} \int_{k_{\min}}^{k_{\max}} k^{3-\beta} e^{-2kh} dk$

At low h , $k^{3-\beta}$ is integrable only when

$$3-\beta > -1 \text{ or when } \beta < 4$$

At $\beta = 4$: $\int \frac{dk}{k}$ has logarithmic divergence.

At very low altitude

$$h \ll k_{\max}^{-1}$$

Intermediate altitude

$$k_{\max}^{-1} \ll h \ll k_{\min}^{-1}$$

Very high altitude

$$h \gg k_{\min}^{-1}$$

(C)(c) Vector magnetometer

The components given by vector magnetometer depend on attitude accuracy.

$$B(r) = B_0 + b(r)$$

where

$$B_0 = \|B_0\| \sim 5 \times 10^4 \text{ nT}$$

For a small attitude error $\delta\theta$,

$$R(\delta\theta) \simeq I - [\delta\theta]$$

first order magnetic error is:

$$\delta B_{\text{att}} \simeq \delta\theta \times B_0$$

Vector and Scalar position information

Let G_V be a characteristic vector-anomaly gradient scale and

G_S corresponding scalar anomaly gradient.

If the intrinsic sensor noise is σ , a vector component has variance

$$\sigma_v^2 = \sigma^2 + B_I^2 \epsilon^2$$

Therefore characteristic vector position information

$$\text{scales as } I_v \sim \frac{G_v^2}{\sigma^2 + B_I^2 \epsilon^2}$$

$$\text{while scalar scales as } I_s \sim \frac{G_s^2}{\sigma^2}$$

Defining the threshold ϵ_* by

$$I_v = I_s$$

then

$$\frac{G_v^2}{\sigma^2 + B_I^2 \epsilon_*^2} = \frac{G_s^2}{\sigma^2}$$

$$\Rightarrow \sigma^2 G_v^2 = G_s^2 (\sigma^2 + B_I^2 \epsilon_*^2)$$

$$B_I^2 \epsilon_*^2 = \sigma^2 \left(\frac{G_v^2}{G_s^2} - 1 \right)$$

$$\therefore \epsilon_{\perp} = \frac{\sigma}{B_{\perp}} \sqrt{\frac{Gv^2}{G_s^2} - 1}$$

Thus the vector measurement is useful only when the vector spatial-gradient information it carries exceeds attitude induced noise penalty.

Let's take,

$$B_{\perp} \epsilon \leq \sigma$$

so,

$$\epsilon \leq \frac{\sigma}{B_0}$$

$$\text{if } B_0 = 50,000 \text{ mT}, \sigma = 2 \text{ nT}$$

$$\epsilon \leq \frac{2}{50,000} = 4 \times 10^{-5} \text{ rad}$$

$\therefore$ attitude accuracy has to be extremely good.